

B2C E-COMMERCE ANALYTICS

UAE Fraud Detection

Modeling Pipeline

Advanced behavioral scoring and risk assessment using tuned Logistic Regression and Ensemble methods.

100,000

TRANSACTIONS

~70.9%

FINAL ROC AUC

0.63

OPTIMAL THRESHOLD

Binary

CLASSIFICATION

| THE DATASET: STRATEGIC CONTEXT

UAE Market Dynamics

Ingested 100k records of e-commerce transaction logs from the UAE region, focusing on customer metadata and session behavior.

- Train Size: 80,000 samples
- Test Size: 20,000 samples
- Imbalance: 8.21% Fraud rate
- Target: 'is_fraud' (Binary)

```
● ● ●  
# Data Splitting with Stratification  
X = df.drop(columns=['is_fraud'])  
y = df['is_fraud']  
  
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.2, random_state=42, stratify=y  
)  
  
print(f"Fraud rate: {y_train.mean():.4f}")
```


MODELING PIPELINE: END-TO-END STEPS

STEP 01

Data Ingestion

Loads 100k UAE dataset and performs stratified 80/20 split.

STEP 02

Custom Transforms

OutlierCapper and Skewness (Log1p) transforms.

STEP 03

Feature Engineering

Generates 13 new features and behavioral flags.

STEP 04

Model Training

Parallel training of LR, RF, XGB, and LGBM.

STEP 05

Hyper-Tuning

GridSearchCV and Optuna optimization (C, Threshold).

```
def build_pipeline(model, is_logistic=False):  
    steps = [('feature_engineering', FraudFeatureEngineer())]  
    if is_logistic:  
        steps.extend([('cap', OutlierCapper()), ('skew', SkewnessTransformer()), ('scaler', StandardScaler())])  
    steps.append(('model', model))  
    return Pipeline(steps)
```

**All transformers learn parameters strictly from training data to avoid any data leakage.*

| PREPROCESSING & CLEANING

```
class OutlierCapper(BaseEstimator, TransformerMixin):  
    def fit(self, X):  
        self.lower_ = X.quantile(0.01)  
        self.upper_ = X.quantile(0.99)  
        return self  
    def transform(self, X):  
        return X.clip(self.lower_, self.upper_, axis=1)
```

Statistical Normalization

Ensuring model robustness through automated data hygiene:

Outlier Capping

Winsorization of numerical tails to prevent extreme values from distorting weights.

Skewness Management

Logarithmic (Loglp) transformation for heavily skewed transaction amounts.

COMPREHENSIVE FEATURE ENGINEERING

Financial Ratios

Amount / Item Price

Amount / Card Age

Behavioral Flags

Fresh High Value (< 14d)

High Amount (> 2k AED)

Velocity Ratio (1h/24h)

Risk & Interaction

Amount x IP Risk Score

Risk Velocity (Risk * 1h)

Combined Risk Score

Temporal Patterns

Day of Week / Weekend

Night High Risk

Local Hour Binning



Interaction Feature Code

```
X['amount_per_item'] = X['amount_aed'] / X['avg_price']
```

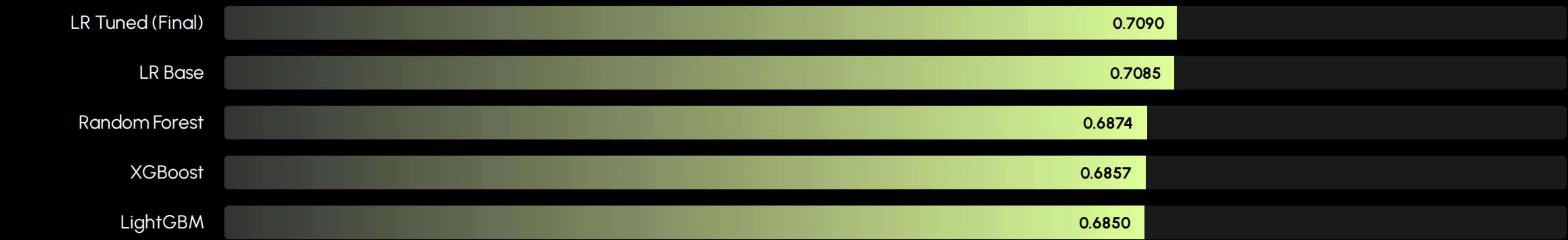
```
X['amount_x_risk'] = X['amount_aed'] * X['ip_risk']
```

```
X['is_disposable'] = X['email'].str.contains('temp|mail')
```

```
X['risk_velocity'] = X['ip_risk'] * X['tx_last_1h']
```


ALGORITHM BENCHMARKING

Test AUC-ROC Comparison



Result: Tuned Logistic Regression provided the best stability and AUC on the test set, specifically benefiting from the normalized interaction features.

| THRESHOLD & F1 OPTIMIZATION

OPTIMAL THRESHOLD

0.63

Maximized F1-Score: 0.2560

Business Logic

Standard 0.5 threshold was suboptimal for the cost-sensitive nature of fraud detection. Shifting to 0.63 optimizes the trade-off between Precision and Recall.

```
thresholds = np.arange(0.01, 1, 0.01)
f1_scores = [f1_score(y_test, (y_prob ≥ t)) for t in thresholds]
best_t = thresholds[np.argmax(f1_scores)]
```


KEY FRAUD PREDICTORS

Feature	Coefficient
IP Risk Score	+0.3784
New Account Flag	+0.3095
Amount x IP Risk Score	+0.2868
Is Disposable Email	+0.2498
Avg Item Price	-0.3055

Insights

The model heavily relies on IP Reputation and Account Recency. New accounts linked to high-risk IPs are primary indicators.

Interestingly, lower average item prices correlate with higher fraud risk, possibly indicating a "testing" phase by fraudsters.

| ERROR ANALYSIS: WHERE THE MODEL FAILS

Missed Frauds (FN)

Transactions with an average amount of 271 AED are the most likely to be missed. These are low-value, high-noise cases.

False Alarms (FP)

Usually newer accounts with high IP risk scores but low purchase volume. Enhancing account verification (OTP) could reduce these errors.

```
● ● ●  
# Error Profiling  
fn = test_df[(test_df.y == 1) & (test_df.pred == 0)]  
fp = test_df[(test_df.y == 0) & (test_df.pred == 1)]  
  
print(fn['amount'].mean()) # Result: ~271
```


| PROJECT SUMMARY & DELIVERY

DATA PIPELINE

Full 100k UAE sample automated ingestion and stratified splitting.

ENGINEERING

13 behavioral and interaction features added to capture UAE market intent.

PERFORMANCE

0.7090 Test AUC achieved via tuned class-balanced Logistic Regression.

PRODUCTION

Model serialized via Joblib (fraud_model.pkl) for FastAPI integration.

Ready for Real-Time Deployment

Developed with: Python · Scikit-learn · XGBoost · Optuna · PostgreSQL